

## Chapter 1 – Introduction to Science

### Invention and discovery 發明與發現



| Term                | Definition                                                                                                                           |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Invention 發明</b> | The creation of a new device, process, or method that has <b>not previously existed.</b><br>創造出一種全新的設備、過程或方法，而之前從未存在過。               |
| <b>Discovery 發現</b> | The act of finding or uncovering something that <b>already exists in nature</b> but was previously unknown.<br>找到或揭示自然界中已存在但之前未知的事物。 |

#### Concept check

|                                                                |                       |
|----------------------------------------------------------------|-----------------------|
| <b>Drones 無人機</b>                                              | Invention / Discovery |
| <b>Electricity 電</b>                                           | Invention / Discovery |
| <b>Gravity 重力</b>                                              | Invention / Discovery |
| <b>Airplane 飛機</b>                                             | Invention / Discovery |
| <b>Telephone 電話</b>                                            | Invention / Discovery |
| <b>Silver 銀</b>                                                | Invention / Discovery |
| <b>Penicillin 盤尼西林 (a chemical found in microorganism 微生物)</b> | Invention / Discovery |
| <b>Light Bulb 電燈泡</b>                                          | Invention / Discovery |
| <b>DNA Structure DNA 結構</b>                                    | Invention / Discovery |
| <b>Internet 互聯網</b>                                            | Invention / Discovery |

## Branches of science 科學分支

|                                                                                                                                                                        |                                                                                                                                                                         |                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Physics 物理學:</b><br/>study of <b>forces and movement</b><br/>研究力和運動</p>         | <p><b>Chemistry 化學:</b><br/>study of <b>reaction</b> of matter<br/>研究物質的反應</p>         | <p><b>Biology 生物學:</b><br/>study of <b>living things</b><br/>研究生物</p>                                  |
| <p><b>Astronomy 天文學:</b><br/>study of <b>stars and the Universe</b><br/>研究星星和宇宙</p>  | <p><b>Geology 地質學:</b><br/>study of <b>earth's inner structure</b><br/>研究地球的內部結構</p>  | <p><b>Medicine 醫學:</b><br/>study of <b>diseases</b> and ways to <b>treat</b> them<br/>研究疾病及其治療方法</p>  |

### Concept check

| Statement                                                                           | Science Branch |
|-------------------------------------------------------------------------------------|----------------|
| The study of how forces act on objects and cause movement.                          |                |
| Experimenting with acids 酸 and alkalis 鹼 to understand pH levels. 酸鹼度。              |                |
| Observing and analyzing the behavior 行為 of living organisms 生物體。                    |                |
| Investigating the formation and evolution 演化 of stars and galaxies 星系。              |                |
| Examining the layers and composition 組成 of Earth's crust 地殼。                        |                |
| Developing vaccines 疫苗 to prevent the spread of viruses. 病毒                         |                |
| Learning about the gravitational pull 引力 of planets 行星 and its effect on orbits 軌道。 |                |
| Analyzing the structure and function of the human body.                             |                |
| Studying the impact of friction 摩擦力 on the motion of objects.                       |                |

## Other types of scientific investigations

### A. Classifying 分類

- Comparing the similarities and differences between objects, and sorting those with similarities into groups



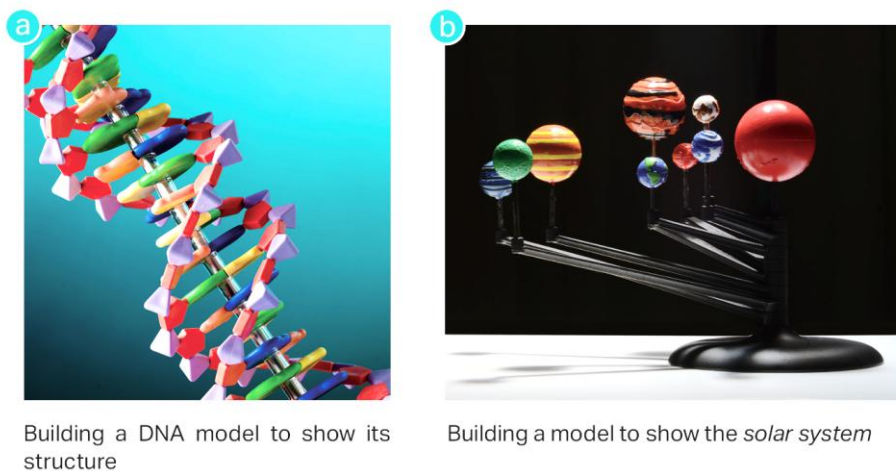
### B. Pattern seeking 尋找規律

- Observing and recording natural phenomena, and trying to find a pattern



### C. Model building 製作模型

- Use models to represent objects that are too small or too big to see



# IS regular course 2025-2026

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| English Term  | Type      |
|---------------|-----------|
| Drones        | Invention |
| Electricity   | Discovery |
| Gravity       | Discovery |
| Airplane      | Invention |
| Telephone     | Invention |
| Silver        | Discovery |
| Penicillin    | Discovery |
| Light Bulb    | Invention |
| DNA Structure | Discovery |
| Internet      | Invention |

## Branches of science 科學分支

| Statement                                                                           | Science Branch |
|-------------------------------------------------------------------------------------|----------------|
| The study of how forces act on objects and cause movement.                          | Physics        |
| Experimenting with acids 酸 and alkalis 鹼 to understand pH levels. 酸鹼度。              | Chemistry      |
| Observing and analyzing the behavior 行為 of living organisms 生物體.                    | Biology        |
| Investigating the formation and evolution 演化 of stars and galaxies 星系.              | Astronomy      |
| Examining the layers and composition 組成 of Earth's crust 地殼.                        | Geology        |
| Developing vaccines 疫苗 to prevent the spread of viruses. 病毒                         | Medicine       |
| Learning about the gravitational pull 引力 of planets 行星 and its effect on orbits 軌道. | Astronomy      |
| Analyzing the structure and function of the human body.                             | Biology        |
| Studying the impact of friction 摩擦力 on the motion of objects.                       | Physics        |